7.8 Finding the Area of Composite Figures

Lesson Introduction

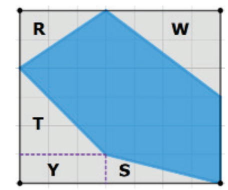
Benchmark	MA.7.GR.1.2 Solve mathematical or real–world problems involving the areas of polygons or composite figures by decomposing them into triangles or quadrilaterals.	Learning Target	• I can decompose composite figures into triangles and quadrilaterals to find the area.

	Building On	Working Toward
Benchmark Coherence	MA.6.GR.2.2 Solve mathematical and real-world problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles.	MA.912.GR.4.4 Solve mathematical and real-world problems involving the area of two-dimensional figures.

Essential Questions	• How do you decompose composite figures into triangles and quadrilaterals to find the area? • How do you find the area of a composite figure inscribed in a rectangle?
Lesson Narrative	Students explore the similarities and differences between decomposing a composite figure to find its area and inscribing that composite figure within a rectangle to find the area in a different way. Students interact with triangles and rectangles as they work to decompose composite figures while making meaning of the purpose and intent of inscribing within a rectangle.

Component Summary	7.8.1 Warm-Up (~5 min.)	Identify math errors in an advertisement *(SE p. 338)*	7.8.4 Guided Practice (10 min.)	Practice decomposing composite figures to find their areas *(SE p. 341)*
	7.8.2 Guided Instruction (20 min.)	Find areas of composite figures by decomposing and by inscribing them in rectangles *(SE p. 339)*	7.8.5 Your Turn! (5 min.)	Find the area of a composite polygon *(SE p. 342)*
	7.8.3 Collaboration (10 min.)	Collaboratively decomposing a composite figure on grid paper to find its area *(SE p. 340)*	7.8.6 Wrap-Up (~5 min.)	Reflect on mastery of lesson learning targets *(SE p. 343)*

	Glossary	Materials	Additional Preparation
Resources	Key Terms: • Composite figure Additional Terms: N/A	• (Optional) Graph paper • (Optional) Scissors • (Optional) Markers	N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>After decomposing the composite figures, students may struggle to find the lengths of sides not given in the triangles or quadrilaterals.</td><td>In 7.8.4 Guided Practice prompt students to create a table with the types of shapes they decompose the figures into. Creating a table (like the one in 7.8.3 Collaboration) will help students organize their work.
Consider allowing students to use calculators.
If time is an issue, then consider assigning 7.8.4 Guided Practice and/or 7.8.5 Your Turn! for homework.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
During 7.8.1 Warm-Up prompt students to initially think about the math fail and write their response independently. Next, prompt students to share their thinking with a partner. Partners can provide each other with feedback before the whole-class discussion.</td><td>MA.K12.MTR.4.1: Discussion and Reflection
Students collaborate with a partner in 7.8.3 Collaboration as they find the areas of composite figures. Students have opportunities to discuss their thinking with their peers. Select and highlight student work that advances and deepens the development of correct and increasingly efficient methods.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">In 7.8.3 Collaboration question 1, students can use a graphic organizer to keep track of similarities and differences, with the potential to use different colors to show their partner’s method of decomposing the figure.
In 7.8.2 Guided Instruction consider prompting students to use graph paper to draw the figure and inscribe it in a rectangle. Students can cut out each shape that is not a part of the composite shape. Then, students can calculate and record the area for each one.
</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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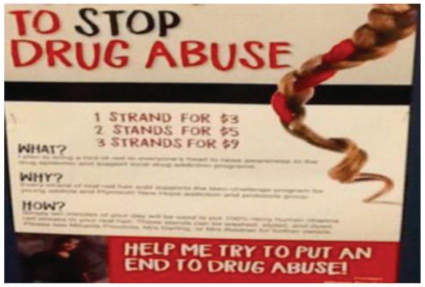
7.8.1 Warm-Up: Math Fail

Component Overview: *Students identify and explain mistakes in a photograph of a sign. Use the Think-Pair-Share routine to engage student thinking and interaction.*

7.8.1 Warm-Up: Math Fail

The advertisement below is for the sale of colored strands for braids to raise money for a Stop Drug Abuse campaign.



1. Explain what the math fail is in the advertisement.
   Answers will vary. The math fail is that the rates are inconsistent; the 5 to 2 rate is not the same as the other two 3 to 1 rates.

Encourage creativity and thinking outside of the box. Students could respond in different ways, and both be 'correct' depending on their justification.

- *Students may identify a* $3:1$ *ratio, stating that two threads should cost* \$6.
- *Students may believe that unit price should decrease as quantity increases to encourage people to by more. So, if buying* 2 *for* \$5 *saves a dollar, then buying* 3 *could cost* \$7 *to save* \$2.

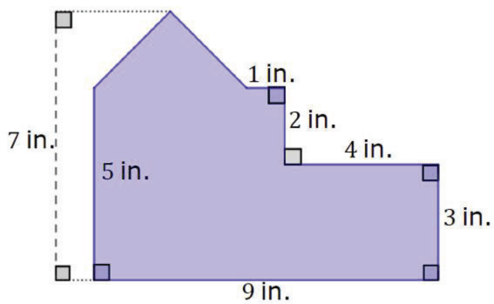
7.8.2 Guided Instruction: Finding the Area of Polygons

Component Overview: *Students decompose composite figures into triangles and quadrilaterals. Then, they determine the dimensions of the triangles and quadrilaterals and find their areas. Next, students calculate the sum of the areas to determine the area of the composite figure. Students also inscribe a composite figure in a rectangle. They will calculate the areas of triangles and quadrilaterals outside of the composite figure. Students subtract the sum of these areas from the area of the rectangle to determine the area of the composite figure.*

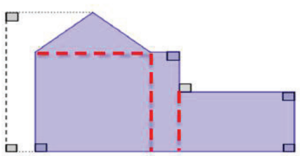
7.8.2 Guided Instruction: Finding the Area of Polygons

Composite figures are two- or three-dimensional figures that can be decomposed into smaller figures.

A floorplan is shown. All segment lengths are in inches (in.).



1. With your partner, decompose the floorplan into triangles and quadrilaterals.
   Answers will vary.



2. Find the area of the floorplan. Show your work.
   $A = \frac{1}{2}bh;\ A = \frac{1}{2}\cdot 4\cdot 2;\ A = \frac{1}{2}\cdot 8;\ A = 4\ in^2$
   $A = bh;\ A = 4\cdot 5;\ A = 20\ in^2$
   $A = bh;\ A = 1\cdot 5;\ A = 5\ in^2$
   $A = bh;\ A = 4\cdot 3;\ A = 12\ in^2$
   Total Area: $4 + 20 + 5 + 12 = 41\ in^2$

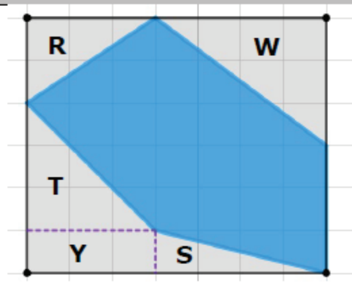
What shapes did you decompose your figure into? How can you use the given dimensions to determine the unknown dimensions?

Think-Pair-Share
Prompt students to initially think about their mistakes and write their responses independently. Next, prompt students to share their thinking with a partner. Partners can provide each other with feedback before the whole-class discussion.

7.8.2 Guided Instruction: Finding the Area of Polygons (continued)

3. A polygon inscribed inside a rectangle is shown on the grid. Each grid line is 1 unit.

   a. Complete the table.

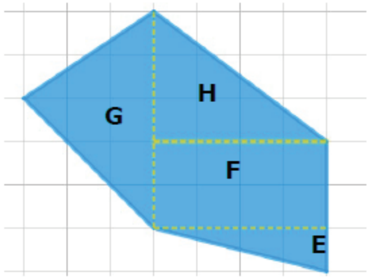
Polygon Inscribed in a Rectangle


Decomposition of Rectangle
Area of the large rectangle $42\ unit^2$
Area of R $3\ unit^2$
Area of S $2\ unit^2$
Area of T $4.5\ unit^2$
Area of W $6\ unit^2$
Area of Y $3\ unit^2$
Area of the blue polygon $23.5\ unit^2$

Consider having students use graph paper to draw and cut out a 6 by 7 unit rectangle. Then, students can draw and cut out R, S, T, Y, and W. This will provide a visual and kinesthetic entry-point for calculating the areas in the decomposition of the rectangle.

7.8.2 Guided Instruction: Finding the Area of Polygons (continued)

b. The same polygon is shown on the grid. Each grid line is 1 unit.

Polygon

Decomposition of Polygon
Area of E $2\ unit^2$
Area of F $8\ unit^2$
Area of G $7.5\ unit^2$
Area of H $6\ unit^2$
Area of the blue polygon $23.5\ unit^2$

c. Discuss with your partner the different possible methods of finding the area of the blue polygon. List one strength of each method.
Answers will vary. One method is to subtract the area surrounding the blue polygon from the area of the large rectangle. A strength is using shapes already defined. Another method is to decompose the blue polygon into various shapes and to find the total area of the shapes. A strength is finding the total area of less shapes.

What are the advantages of inscribing a composite figure into a rectangle to find area? What are the disadvantages? Which method do you prefer to use?

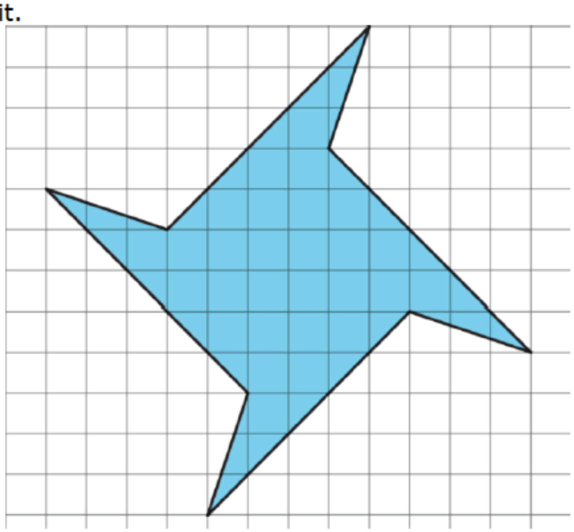
Think-Pair-Share
Prompt students to initially think about their mistakes and to write their responses independently. Next, prompt students to share their thinking with a partner. Partners can share feedback with each other before the whole-class discussion.

7.8.3 Collaboration: Area of a Polygon

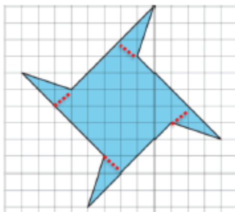
Component Overview: *Students collaborate to find various ways to decompose a composite figure, which is in the shape of a spinner. The students will find the area of each shape, and then find the total area of the composite figure.*

7.8.3 Collaboration: Area of a Polygon

1. A polygon is shown on grid paper. Each grid line is 1 unit.



   a. Decompose the polygon into triangles and quadrilaterals.



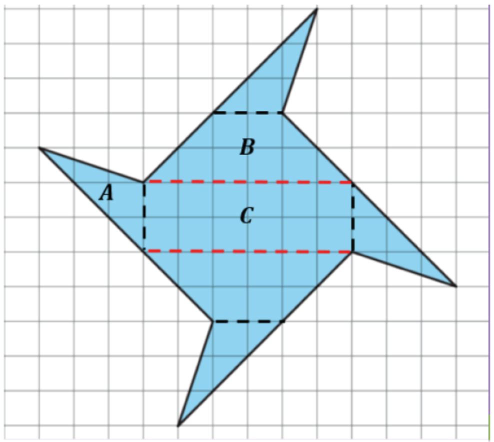
   b. Share your decomposition with your partner. List at least one similarity and one difference.
   Answers will vary. Ours is similar because we decomposed it into the same amount of polygons. Ours is different because we decomposed it into a different amount of triangles and quadrilaterals.

Provide colored pencils for students to sketch on their Workbook for support in decomposing, or provide copies of the shape on the coordinate plane for students to cut.

7.8.3 Collaboration: Area of a Polygon (continued)

2. Taahira decomposed the polygon into 7 pieces using 3 different shapes, as shown. Each grid line is 1 unit.

Work with your partner to find the area of each shape and the total area of the polygon.



Consider having students decompose to find the total area while also inscribing within a rectangle to verify their answers.
Students can work in pairs, with one student decomposing like Taahira while the partner is inscribing to verify.
Students could also choose their own method with a partner before finding someone around the room who did it a different way to verify if they got the same answers.

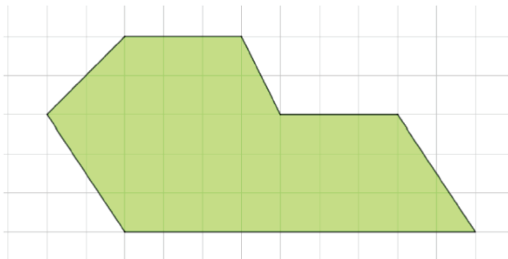
Shape	Type of Shape	Area of the shape	Number of Shapes	Total Area
A	triangle	$3\ u^2$	4	$12\ u^2$
B	trapezoid	$8\ u^2$	2	$16\ u^2$
C	rectangle	$12\ u^2$	1	12
Area of the Polygon				$40\ u^2$

7.8.4 Guided Practice: Decomposing Composite Figures to Find Area

Component Overview: *Students practice finding the areas of composite shapes. For question 1, students determine the dimensions by counting units on a grid. The second figure is not drawn on a grid, but the necessary dimensions are provided.*

7.8.4 Guided Practice: Decomposing Composite Figures to Find Area

1. A polygon is shown on a grid. Each grid line is 1 unit.

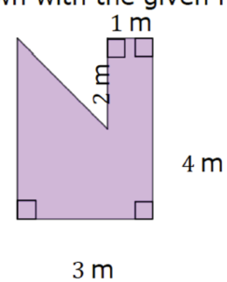


Find the area of the polygon.
$36\ u^2$

Encourage students to draw either:
- *the decomposed triangles and quadrilaterals*
- *the rectangle in which the figure is inscribed*

Prompt students to label the dimensions and make a table to organize the areas of the triangles and quadrilaterals.

2. A polygon is shown with the given measurements in meters (m).



What is the area of the composite figure?
$10\ m^2$

What method did you use to determine the area? How can you use the given dimensions to determine the unknown dimensions?

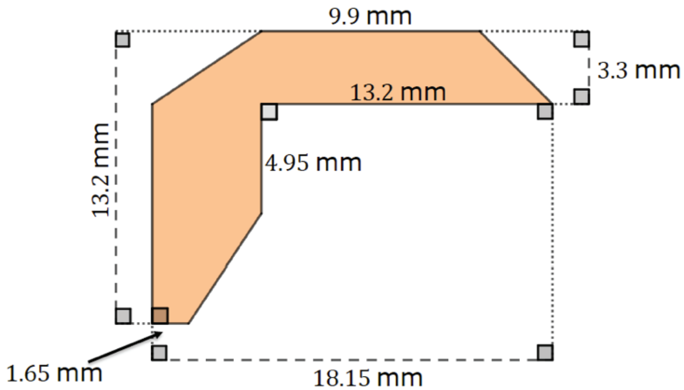
Consider challenging students who finish early to find the areas of the two figures using a different approach. Prompt students to recognize that this is a way to check their work.

7.8.5 Your Turn!

Component Overview: *Students practice independently finding the area of a composite figure. Consider providing students with graph paper and scissors so they can physically model the problem if they wish.*

7.8.5 Your Turn!

1. Find the area of the polygon. Lengths are shown in millimeters (mm).



$87.12\ mm^2$

Consider challenging students who finish early to draw and label their own composite figures. Then, prompt students to swap figures with a partner and calculate the area of their partner's figure.

7.8.6 Wrap-Up: Reflection

Component Overview: *Students reflect on mastery of lesson learning targets. Consider using data from this formative assessment to identify and address any student misunderstandings.*

7.8.6 Wrap-Up: Reflection

1. Place an **X** on the number in the emoji scale that indicates your understanding in relation to today's learning target.
   Answers will vary.

   I can decompose composite figures into triangles and quadrilaterals to find the area.

2. Provide evidence to explain your self-assessment.
   Answers will vary.

Consider creating a checklist designed for your learning targets. Students can create an emoji for each item or learning target.
I can decompose composite figures into triangles and quadrilaterals.
I can label the decomposed figures with the correct measures.
I can apply the correct formulas to find the area for each decomposed figure.
I can correctly calculate rational numbers to find the area.